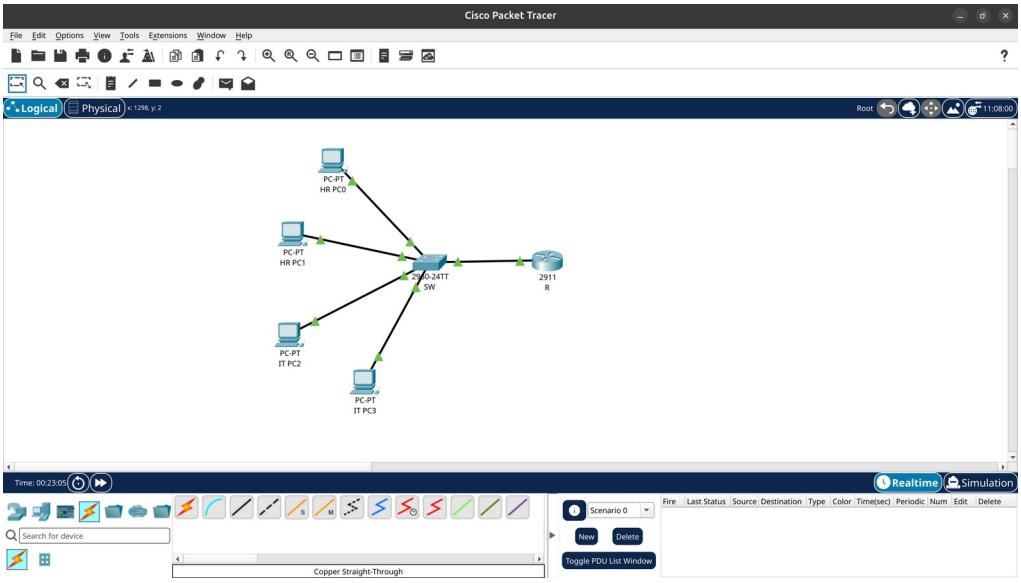
This is my first project....

Goal: Building an office with two departments, HR and IT.
Each department gets its own network.
They are separated by VLANs, but the router lets them talk to each other.
PCs get their IP addresses automatically(DHCP).

To build this i use an app called packet tracer and it is free you can download and use it.
i use 1 switch, 1 router, and 4 pc's(2 for HR and 2 for IT) to build this project.

The topology



After this we have to configure our switch

enable
configure terminal
hostname SW1

vlan 10
name HR
vlan 20
name IT
exit

interface range fa0/2 - 3
switchport mode access
switchport access vlan 10

interface range fa0/4 - 5
switchport mode access
switchport access vlan 20
exit

```
interface fa0/1
switchport mode trunk
exit

end
write memory
```

and configure our router

```
enable
configure terminal
hostname R1

interface gigabitEthernet 0/0
no shutdown
exit

interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

configure DHCP on the router

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10

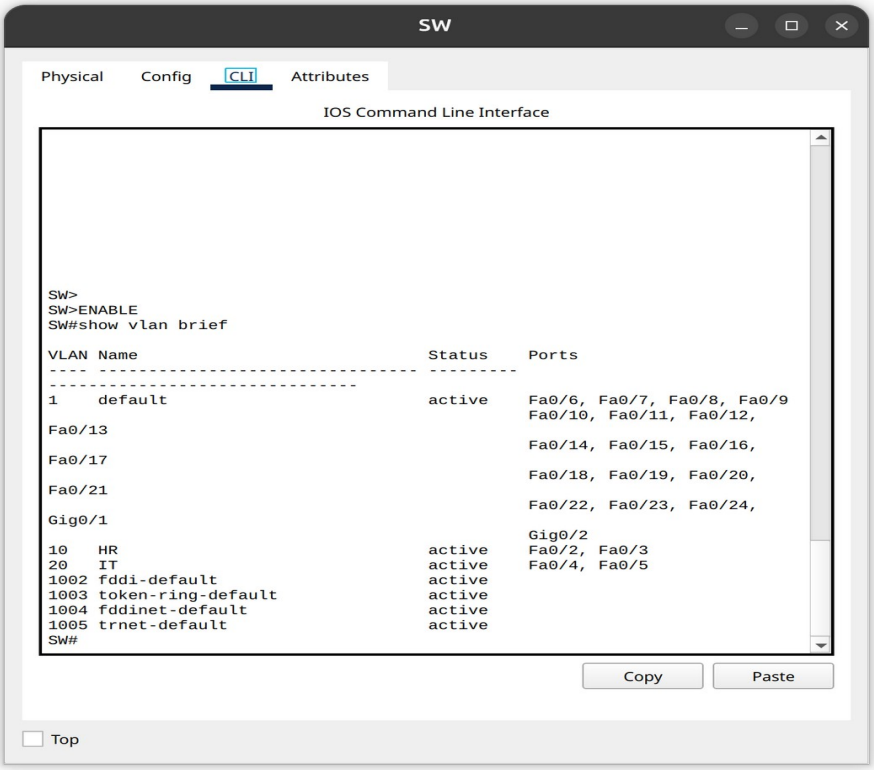
ip dhcp pool HR
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit

ip dhcp pool IT
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit

end
write memory
```

and then set the pc’s to use DHCP

OUTPUT OF SHOW VLAN BRIEF



THE PINGING ALSO WORKS

